

Notes: Agarwal, Mukherjee & Naaraayanan (RFS, 2023)

Roads and Loans

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1 Big picture

- **Question.** When productive opportunities improve for poor rural households, does private bank finance actually respond?
- **Setting.** The paper studies India's rural road program, Pradhan Mantri Gram Sadak Yojana (PMGSY), and asks whether new road access brings formal bank credit to villagers.
- **Identification.** PMGSY prioritized villages above population thresholds. In the states studied, the relevant thresholds are 500 and 1,000 people. Villages just above and just below these cutoffs are compared.
- **Data.** The core micro data are proprietary loan-level records from a large Indian bank operating in selected villages in Odisha and Uttarakhand. The authors also use village census data, road-treatment data, and district-level credit and GDP data.
- **Main extensive-margin result.** New roads raise the likelihood that the bank lends to a village and raise the number of villagers receiving loans. The lower estimate implies about 75% more borrowers in above-cutoff villages.
- **Main intensive-margin result.** Conditional on receiving loans, average lending amounts are about 30–35% higher in villages just above the road-priority threshold.
- **Loan quality and pricing.** Loan maturities, default outcomes, and interest rates are similar across above- and below-cutoff villages. Roads increase credit quantity, not riskier pricing or worse repayment.
- **Mechanism evidence.** The extra lending is concentrated in productive loans rather than consumption or social-purpose loans. The evidence is more consistent with a demand-side response to higher productive opportunities than with banks simply finding it cheaper to supply credit.
- **Distributional result.** Low-asset borrowers benefit disproportionately. This suggests roads and finance can relax constraints for poorer households rather than only raising collateral values for richer households.
- **Macro result.** At the district level, recent PMGSY road construction is associated with more rural private bank credit and deposits. Roads are associated with stronger output growth in districts with deeper rural credit markets.
- **Economic takeaway.** Infrastructure and finance are complements. Roads can create profitable opportunities, but formal credit helps rural households convert those opportunities into borrowing, investment, and growth.

2 Data and measurement

2.1 The PMGSY road program

- PMGSY was designed to connect previously unconnected habitations to all-weather roads.
- The program used population-based priority rules. Larger unconnected villages were supposed to receive roads before smaller ones.
- The paper focuses on discontinuities around population thresholds. Villages just above the threshold were more likely to receive a road than villages just below it.
- The authors use villages that were unconnected at the start of the sample and use 2001 census population, measured before PMGSY implementation, as the running variable.

Interpretation. Using predetermined census population reduces the concern that village population was manipulated in response to the program.

2.2 States and sample

- The main bank-lending analysis is in Odisha and Uttarakhand.
- These states provide enough observations around the 500 and 1,000 population thresholds and are within the bank’s operating area.
- The bank-lending sample contains 58 villages in the authors’ core geography.
- Because the loan sample is small, the main lending tests combine the 500 and 1,000 cutoffs using normalized population.

Interpretation. The small bank sample means the paper is careful to frame the evidence as a quasi-experimental treatment-control design around the policy cutoffs.

2.3 Bank loan data

- The loan-level data come from one of India’s largest publicly traded banks.
- The sample includes individuals who had a record with the bank by the end of 2014.
- Loan variables include amount disbursed, maturity, interest rate, overdue amount, borrower age, gender, schooling, income, assets, land ownership, and loan purpose.
- The average loan amount is about Rs. 28,000, or roughly US\$480 at the exchange rate used in the paper.
- Average maturity is around three years, and average interest rates are about 14.7%.

Interpretation. The proprietary data let the authors observe actual lending quantities, prices, terms, and repayment outcomes, rather than only aggregate credit volumes.

2.4 Outcome variables

The main micro outcomes are:

- **Bank entry:** an indicator that the bank lent to at least one villager in the village.
- **Number of customers:** $\log(1 + \text{borrowers})$ from a village.
- **Loan amount:** net disbursement scaled by annual household income.
- **Loan quality:** maturity, overdue amount, and percentage overdue.
- **Interest rate:** the average interest rate on the borrower’s loans.
- **Loan type:** productive loans versus nonproductive loans.

Interpretation. Scaling loan amounts by household income makes the economic magnitude comparable across borrowers.

2.5 Macro data

- The paper also studies district-level private bank outcomes from 2004 to 2012.
- District-level bank outcomes include rural credit, rural branches, rural deposits, and corresponding urban outcomes.
- The main road variable is the aggregate length of PMGSY roads constructed in the past three years.
- District-level growth outcomes use GDP measures overall, in agriculture, and in services.

Interpretation. The macro tests ask whether the micro evidence generalizes beyond one bank and whether finance mediates the growth impact of roads.

3 Models and empirical specifications

3.1 First stage: do population thresholds predict road construction?

The first-stage specification is:

$$\begin{aligned} Roads_{s,v} = & \beta_0 + \beta_1 \mathbf{1}[\text{Population above threshold}] \\ & + \beta_2 \mathbf{1}[500 - h \leq pop_{s,v} < 500 + h] \times (pop_{s,v} - 500) \\ & + \beta_3 \mathbf{1}[1,000 - h \leq pop_{s,v} < 1,000 + h] \\ & + \beta_4 \mathbf{1}[1,000 - h \leq pop_{s,v} < 1,000 + h] \times (pop_{s,v} - 1,000) + \sigma_s + \epsilon_{s,v}. \end{aligned} \quad (1)$$

- $Roads_{s,v}$ equals one if village v in state s receives a PMGSY road by 2014.
- $pop_{s,v}$ is 2001 census population.
- σ_s are state fixed effects.
- h is the bandwidth around the population thresholds.

Interpretation. The coefficient β_1 measures the jump in road-treatment probability at the threshold.

3.2 Reduced-form lending specification

The bank-lending specification is:

$$\begin{aligned} Y_{s,v} = & \gamma_0 + \gamma_1 \mathbf{1}[\text{Population above threshold}] \\ & + \gamma_2 \mathbf{1}[500 - h \leq pop_{s,v} < 500 + h] \times (pop_{s,v} - 500) \\ & + \gamma_3 \mathbf{1}[1,000 - h \leq pop_{s,v} < 1,000 + h] \\ & + \gamma_4 \mathbf{1}[1,000 - h \leq pop_{s,v} < 1,000 + h] \times (pop_{s,v} - 1,000) + \sigma_s + \epsilon_{s,v}. \end{aligned} \quad (2)$$

- $Y_{s,v}$ is a bank lending outcome.
- The main coefficient is γ_1 , the discontinuity in the lending outcome at the population cutoff.
- The paper reports estimates for bandwidths of 200 and 250 around the cutoffs.
- Standard errors are bootstrapped; robustness checks use alternative methods.

Interpretation. This is a reduced-form estimate of the effect of being just above the road-priority threshold. Because the threshold predicts roads, it captures road-induced changes in lending.

3.3 Why the paper does not rely on treatment timing

- The authors do not exploit the exact timing of road construction in the micro lending tests.
- PMGSY rules specify that larger villages are prioritized before smaller villages, but not the precise order among villages near each other.
- The authors therefore take a cross-sectional snapshot at the last available year and exploit discontinuities around the population cutoff.

Interpretation. This avoids relying on possibly endogenous timing of individual village road construction.

3.4 Loan use and mechanism specification

For mechanisms, the same reduced-form design is estimated separately for productive and nonproductive loans:

$$Y_{s,v}^{Productive}, \quad Y_{s,v}^{Nonproductive}.$$

- Productive loans include crop loans, micro-enterprise loans, business expansion, asset acquisition, and working capital.
- Nonproductive loans include consumption needs, marriage, festival expenses, and similar uses.

Interpretation. If roads raise productive opportunities, the increase in borrowing should be concentrated in productive loans.

3.5 Distributional heterogeneity

The heterogeneous-effects specification adds borrower characteristics and interactions:

$$Y_i = \alpha + \theta AboveCutoff_v + X_i' \beta + (AboveCutoff_v \times X_i)' \delta + \text{fixed effects} + \epsilon_i.$$

Key interactions include age, low assets, schooling, caste category, and female borrower.

Interpretation. The interaction terms ask who benefits from the new lending opportunities created by roads.

3.6 Macro credit and growth specifications

At the district-year level, the paper estimates:

$$\Delta \log(1 + Credit_{d,t}) = \alpha_d + \lambda_{s,t} + \beta \log(SumRoad_{d,t-3:t-1}) + \Gamma X_{d,t} + \epsilon_{d,t}.$$

For growth, it estimates interactions between road construction and rural credit depth:

$$\begin{aligned} \Delta \log(GDP_{d,t}) = & \alpha_d + \lambda_{s,t} + \beta_H \log(SumRoad_{d,t-3:t-1}) \times HighCredit_d \\ & + \beta_L \log(SumRoad_{d,t-3:t-1}) \times LowCredit_d + \Gamma X_{d,t} + \epsilon_{d,t}. \end{aligned} \quad (3)$$

Interpretation. The macro tests ask whether roads lead to more private bank activity, and whether districts with deeper rural credit markets convert roads into stronger growth.

4 Identification and empirical design

4.1 What identifies the main effect

- The key identification comes from population cutoffs used by PMGSY.
- Villages just above and just below the population threshold should be similar, except that above-threshold villages were more likely to receive roads.
- The main tests use normalized population and local linear controls around the thresholds.
- State and threshold fixed effects absorb broad regional and cutoff-level differences.

Interpretation. The identifying assumption is local comparability: near the thresholds, crossing the cutoff is as good as random with respect to other determinants of bank lending.

4.2 Evidence against manipulation

- Figure 1 shows no obvious bunching in population around the 500 and 1,000 thresholds.
- Figure 2 implements a McCrary density test and does not reject continuity of the running variable at the cutoff.

Interpretation. If local officials manipulated population counts to get roads, the discontinuity design would be threatened. The density evidence supports validity.

4.3 Evidence on covariate balance

- Figure 4 plots baseline village characteristics around the cutoff.
- Characteristics include schools, health centers, electricity, telegraph office, distance to nearest town, SC/ST share, and irrigated land.
- These do not show strong discontinuities at the cutoff.

Interpretation. Balance supports the claim that the discontinuity primarily affects road treatment, not other village attributes.

4.4 First-stage strength

- Figure 5 shows a visible jump in PMGSY road receipt around the cutoff.
- Table 1 estimates a 6.6–6.8 percentage point increase in the probability of receiving a road by 2014.
- Relative to an 11–12% control mean, this is roughly a 57–60% jump.

Interpretation. The first stage is meaningful: the cutoff changes road access enough to study downstream credit responses.

4.5 Why demand is the favored mechanism

- More people get loans and average loan amounts rise.
- Interest rates and loan maturities do not change significantly.
- Loan performance does not deteriorate.
- New borrowing is concentrated in productive loans.
- The bank told the authors that local rural loan supply is tiny relative to the bank's balance sheet, making a binding supply constraint less plausible.

Interpretation. The evidence is most consistent with roads creating profitable investment opportunities that raise demand for formal credit.

4.6 What the paper cannot fully prove

- The micro loan data come from one bank and two states.
- The bank data cannot show whether formal credit replaced informal moneylender borrowing.
- The micro design is reduced form around population cutoffs rather than a direct IV estimate of loan responses to actual road completion.
- The small village sample limits power for some mechanism and heterogeneity tests.

Interpretation. The paper’s strength is clean quasi-random variation and unusually rich loan data; its main limitation is external validity and sample size.

5 Tables and figures

The figures and tables below are directly cropped from the original paper. Adjacent visuals are grouped to save space and make comparisons faster.

5.1 Figures 1–2: manipulation tests

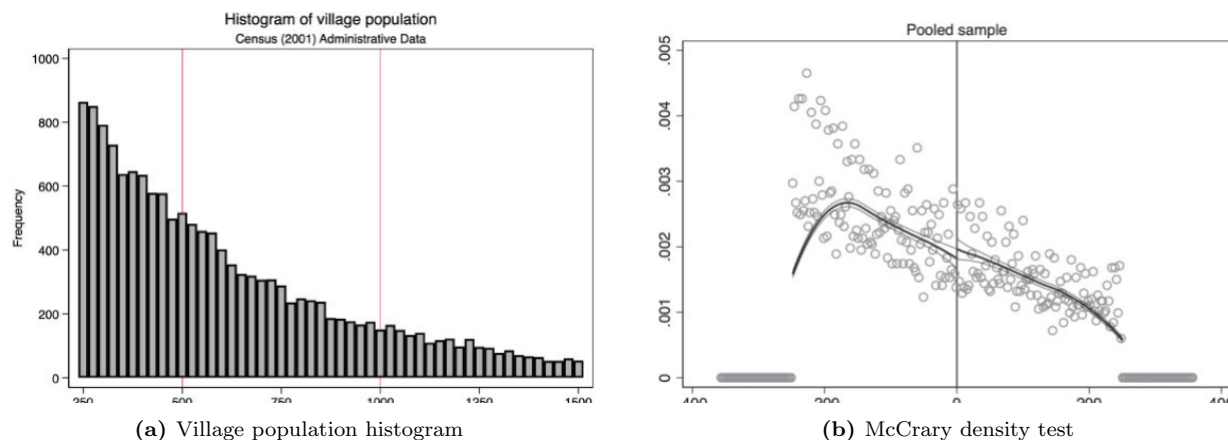


Figure 1: Manipulation tests around PMGSY population thresholds.

Read:

- Figure 1 shows the population distribution around the PMGSY thresholds of 500 and 1,000.
- Figure 2 formally tests for discontinuity in the running variable.
- The McCrary point estimate is small and statistically insignificant.

Economic message. The running variable does not appear manipulated. This supports the discontinuity design.

5.2 Figures 3 and 6: geography of treatment and bank villages

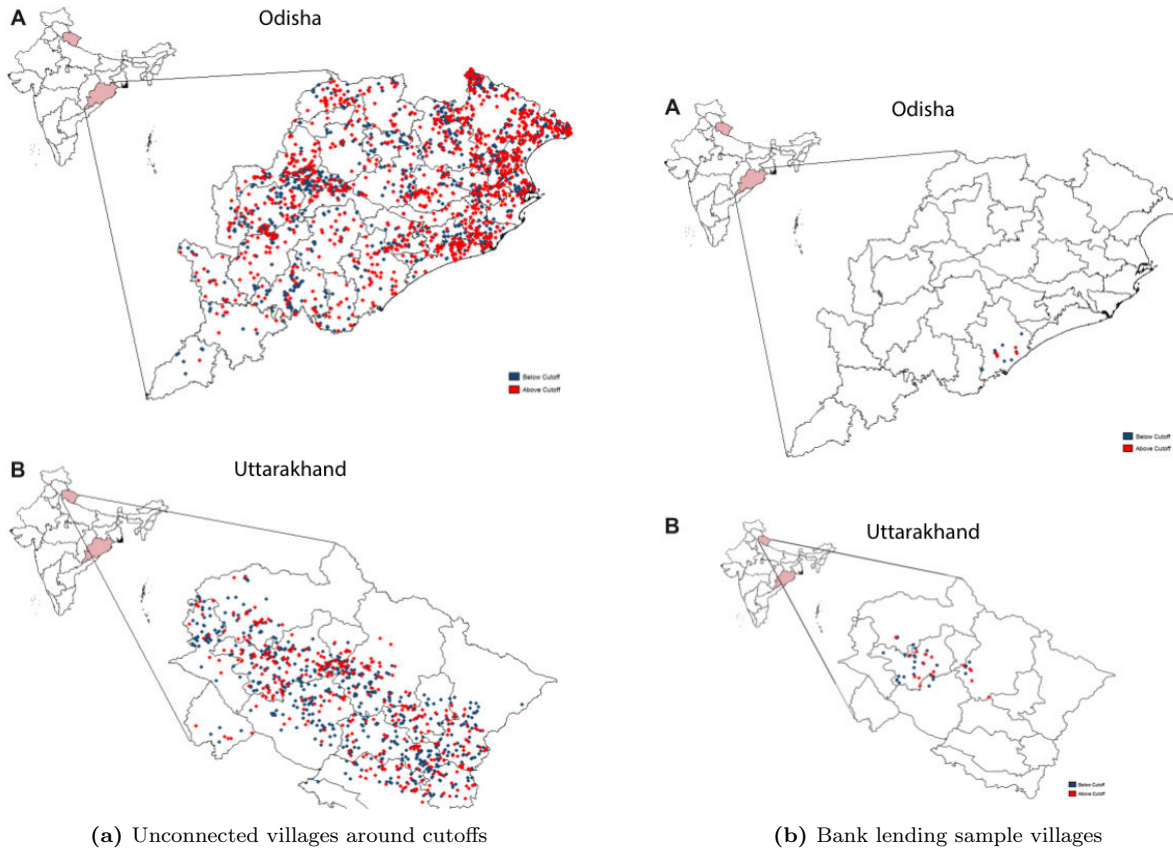


Figure 2: Geographic distribution of treatment and bank sample villages.

Read:

- Figure 3 shows above- and below-cutoff villages in Odisha and Uttarakhand.
- The red and blue villages are geographically interspersed rather than concentrated in separate areas.
- Figure 6 shows the smaller bank-lending sample, again with above- and below-cutoff villages located near each other.

Economic message. The maps support local comparability. Treated and control villages are not visibly separated into different economic regions.

5.3 Figures 4–5 and Table 1: covariate balance and first stage

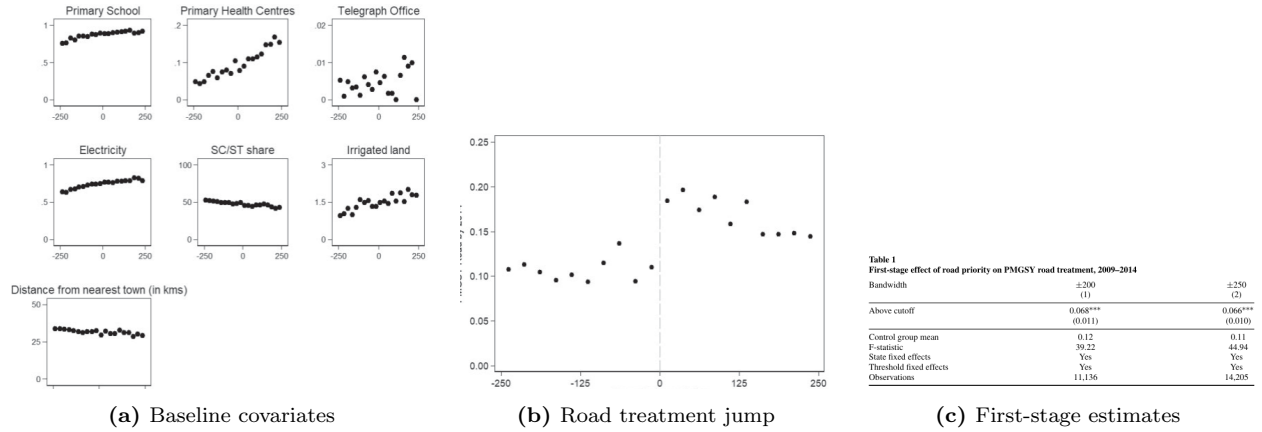


Figure 3: Covariate balance and first-stage evidence.

Read:

- Figure 4 shows no sharp jumps in baseline amenities or demographics around the cutoff.
- Figure 5 shows a clear increase in the probability of receiving a PMGSY road just above the cutoff.
- Table 1 estimates an above-cutoff effect of 0.068 using bandwidth 200 and 0.066 using bandwidth 250.
- The F-statistics are 39.22 and 44.94.

Economic message. The cutoff predicts roads but not other village characteristics. This is the key identification foundation.

5.4 Table 2: bank sample summary statistics

Bandwidth	± 200		± 250	
	(1) Mean	(2) SD	(3) Mean	(4) SD
<i>A. Loan characteristics</i>				
Net disbursement (Rs.)	28,091	19,106	28,064	19,589
Loan maturity (years)	3.06	0.15	3.06	0.15
Interest rate (%)	14.7	3.7	14.7	3.8
Overdue amount (%)	0.12	2.25	0.12	2.19
<i>B. Borrower characteristics</i>				
Age (years)	37	9	37	9
Female (%)	25	43	25	43
Schooling (%)	87	33	87	34
Annual income (Rs.)	131,892	88,718	132,286	89,368

Figure 4: Summary statistics: bank loan sample.

Read:

- Average net disbursement is about Rs. 28,000.
- Average maturity is about 3.06 years.
- Average interest rate is 14.7%.
- Defaults are rare: overdue amount is around 0.12% of loans.
- Borrowers are on average 37 years old; 25% are female; 87% have some schooling.

Economic message. The borrowers are low-income rural households, but the loans are not tiny short-term emergency loans. They are meaningful formal credit contracts.

5.5 Tables 3–4: extensive and intensive margins of lending

Table 3
Impact of new roads on lending: Extensive margin

Bandwidth	± 200 (1)	± 250 (2)
<i>A. Bank entry</i>		
Above cutoff	2.003* (1.059)	1.733** (0.858)
Control group mean	0.439	0.440
State fixed effects	Yes	Yes
Threshold fixed effects	Yes	Yes
Observations	93	116
<i>B. Number of customers</i>		
Above cutoff	0.938** (0.409)	0.652* (0.374)
Control group mean	1.012	0.867
State fixed effects	Yes	Yes
Threshold fixed effects	Yes	Yes
Observations	93	116

The table presents reduced-form estimates of the effect of new rural roads on the propensity of the bank to enter

(a) Extensive margin

Table 4
Impact of new roads on the lending quantities

Bandwidth	±200		±250	
	(1)	(2)	(3)	(4)
Above cutoff	0.025** (0.012)	0.025** (0.012)	0.025** (0.011)	0.030*** (0.011)
Age (years)		−0.001** (0.000)		−0.000* (0.000)
Land		0.006 (0.007)		0.003 (0.007)
log(1+assets)		0.002* (0.001)		0.002** (0.001)
School education		0.009 (0.007)		0.009 (0.006)
Female		−0.051*** (0.006)		−0.049*** (0.006)
Control group mean	0.083	0.083	0.085	0.085
State fixed effects	Yes	Yes	Yes	Yes
Threshold fixed effects	Yes	Yes	Yes	Yes
Observations	1,032	1,032	1,084	1,084

The table presents reduced-form estimates from Equation (2) of the effect of new rural roads on lending activity

(b) Loan quantities

Figure 5: Road priority and bank lending quantities.

Read:

- Table 3 shows that above-cutoff villages are more likely to have bank entry. Odds ratios are 2.003 and 1.733.
- The number of customers increases by 0.938 using bandwidth 200 and 0.652 using bandwidth 250.
- The lower estimate implies around 75% more borrowers relative to below-cutoff villages.
- Table 4 shows that loan amounts scaled by income rise by 0.025–0.030.
- Relative to control means around 0.083–0.085, this is about a 30–35% increase.

Economic message. Roads raise both the number of borrowers and the amount lent to each borrower.

5.6 Tables 5–6: loan maturity, repayment, and interest rates

Table 5
Impact of new roads on loan maturity and quality

Bandwidth	±200			±250		
	log maturity (1)	overdue amount (2)	% overdue (3)	log maturity (4)	overdue amount (5)	% overdue (6)
Above cutoff	-0.009 (0.020)	-164.872 (211.679)	0.057 (0.355)	-0.028 (0.019)	-184.306 (190.637)	-0.066 (0.361)
Control group mean	1.11	104.7	0.12	1.11	100.6	0.12
Loan-purpose fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
State fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Threshold fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	630	630	630	665	665	665

The table presents reduced-form estimates from Equation (2) of the effect of new rural roads on quality of

(a) Maturity and quality

Table 6
Impact of new roads on interest rates

Bandwidth	± 200 (1)	± 250 (2)
Above cutoff	-0.002 (0.006)	-0.005 (0.005)
Control group mean	0.15	0.15
Loan-purpose fixed effects	Yes	Yes
State fixed effects	Yes	Yes
Threshold fixed effects	Yes	Yes
Observations	630	665

(b) Interest rates

Figure 6: Loan maturity, quality, and pricing.

Read:

- Table 5 shows no significant increase in maturity or overdue amounts.
- Coefficients on overdue outcomes are economically small relative to very low default rates.
- Table 6 shows interest-rate effects of -0.002 and -0.005, both insignificant.

Economic message. The increase in credit quantities is not accompanied by worse loan performance or higher interest rates.

5.7 Tables 7–8: falsification and loan-use mechanisms

Table 7
Falsification test: Already connected villages

Bandwidth	±200 (1)	±250 (2)
Above cutoff	0.004 (0.009)	-0.005 (0.008)
Control group mean	0.064	0.072
State fixed effects	Yes	Yes
Threshold fixed effects	Yes	Yes
Observations	2,256	2,675

The table presents reduced-form estimates of the effect of population thresholds on loan disbursed on a placebo

(a) Placebo: already connected villages

Table 8
Impact of new roads on loan types

Bandwidth	Productive loans		Nonproductive loans	
	±200 (1)	±250 (2)	±200 (3)	±250 (4)
Above cutoff	0.043*** (0.011)	0.045*** (0.010)	-0.044*** (0.011)	-0.038*** (0.009)
Control group mean	0.047	0.047	0.066	0.067
State fixed effects	Yes	Yes	Yes	Yes
Threshold fixed effects	Yes	Yes	Yes	Yes
Observations	1,032	1,084	1,032	1,084

This table presents reduced-form estimates quantifying the effect of new rural roads on lending activity based on the type of loan disbursed. Columns 1 and 3 present reduced-form estimates for villages within 200 of the

(b) Productive vs nonproductive loans

Figure 7: Falsification and loan-use mechanisms.

Read:

- Table 7 uses already connected villages as a falsification sample.
- The above-cutoff coefficients are near zero and insignificant.
- Table 8 shows productive loans rise by 0.043–0.045.
- Nonproductive loans fall by 0.038–0.044.

Economic message. The placebo strengthens identification, and the loan-use split supports the productive-opportunity mechanism.

5.8 Tables 9–10: supply-side information and distributional effects

Table 9
Testing for the variability of loan contract terms

	Below threshold (1)	Above threshold (2)	Difference of means (1) - (2)	<i>p</i> -value on difference
<i>A. Bandwidth, ± 200</i>				
Loan amount	0.619	0.635	−0.016	.894
Interest rate	0.273	0.217	0.056	.704
Loan maturity	0.103	0.095	0.008	.869
<i>B. Bandwidth, ± 250</i>				
Loan amount	0.605	0.682	−0.078	.512
Interest rate	0.249	0.249	0.001	.997
Loan maturity	0.092	0.098	−0.006	.895

The table presents results that test the variability of loan contract terms between similar groups of borrowers in above- and below-cutoff villages. We divide our sample into groups of borrowers within above- and below-

(a) Variability of contract terms

Table 10
Heterogeneous effects of new rural roads, by borrower characteristics

Bandwidth	± 200			± 250		
	Loan amount (1)	% Overdue amount (2)	Interest rate (3)	Loan amount (4)	% Overdue amount (5)	Interest rate (6)
Above cutoff	−0.020 (0.026)	0.664 (1.344)	0.012 (0.016)	−0.013 (0.025)	0.555 (1.240)	0.007 (0.016)
Age (years)	−0.001** (0.000)	−0.000 (0.004)	−0.000 (0.000)	−0.000 (0.000)	0.002 (0.004)	−0.000 (0.000)
Low assets	0.004 (0.007)	0.244 (0.243)	−0.000 (0.003)	0.004 (0.007)	0.228 (0.237)	−0.000 (0.003)
School education	−0.006 (0.007)	−0.921 (0.798)	0.000 (0.004)	−0.004 (0.006)	−0.811 (0.726)	−0.001 (0.004)
SC/ST/OBC	−0.002 (0.008)	0.166 (0.186)	0.000 (0.003)	−0.002 (0.008)	0.146 (0.170)	0.000 (0.003)
Female	−0.015*** (0.005)	0.282 (0.280)	0.006* (0.003)	−0.016*** (0.005)	0.248 (0.262)	0.005* (0.003)
Above cutoff x Age (years)	0.001 (0.000)	−0.019 (0.021)	−0.000 (0.000)	0.000 (0.000)	−0.020 (0.020)	−0.000 (0.000)
Above cutoff x Low assets	0.021** (0.010)	−0.206 (0.233)	−0.005 (0.004)	0.020** (0.010)	−0.209 (0.213)	−0.004 (0.004)
Above cutoff x School education	0.015 (0.011)	0.902 (0.788)	0.007 (0.009)	0.012 (0.010)	0.750 (0.697)	0.008 (0.008)
Above cutoff x SC/ST/OBC	−0.011 (0.011)	−0.563 (0.393)	−0.006 (0.005)	−0.008 (0.010)	−0.508 (0.362)	−0.007 (0.005)
Above cutoff x Female	0.001 (0.010)	−0.464 (0.345)	−0.009* (0.005)	0.002 (0.010)	−0.439 (0.317)	−0.008 (0.005)
Control group mean	0.083	0.12	0.15	0.085	0.12	0.15
Loan-purpose fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
State fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Threshold fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,032	630	630	1,084	665	665

(b) Borrower-characteristic heterogeneity

Figure 8: Information and distributional effects.

Read:

- Table 9 tests whether contract-term dispersion differs across above- and below-cutoff villages.
- Differences in variability for loan amount, interest rate, and maturity are small and insignificant.
- This weakens the idea that the main channel is better bank information or screening.
- Table 10 shows that the interaction of above-cutoff with low assets is positive and significant for loan amounts.
- Female borrowers receive lower baseline loan amounts, but the road effect is not mainly a female-borrower effect.

Economic message. Roads appear to help low-asset borrowers disproportionately, consistent with finance reaching constrained households.

5.9 Tables 11–12: macro credit and growth

Table 11
Macro evidence: New rural roads and bank credit

Dependent variable	$\Delta \log (1+\text{rural} \dots)$			$\Delta \log (1+\text{urban} \dots)$		
	Credit (1)	Branches (2)	Deposit (3)	Credit (4)	Branches (5)	Deposit (6)
<i>A. Without controls</i>						
$\log(\text{Sum road}_{t-3,t-1})$	0.466* (0.273)	-0.071 (0.223)	0.705** (0.330)	0.192 (0.561)	-0.058 (0.242)	0.105 (0.459)
Mean of dep. var.	8.86	6.56	7.84	12.7	11.2	11.8
Controls	Yes	Yes	Yes	Yes	Yes	Yes
District fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
State $\times$ year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,524	2,524	2,524	2,524	2,524	2,524
<i>B. With controls</i>						
$\log(\text{Sum road}_{t-3,t-1})$	0.465* (0.269)	-0.088 (0.224)	0.684** (0.331)	0.207 (0.570)	-0.059 (0.242)	0.206 (0.465)
Mean of dep. var.	8.86	6.56	7.84	12.7	11.2	11.8
Controls	Yes	Yes	Yes	Yes	Yes	Yes
District fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
State $\times$ year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,524	2,524	2,524	2,524	2,524	2,524

The table examines the relationship between new rural roads on private sector bank credit extended for the period

(a) Rural roads and bank credit

Table 12
Macro evidence: New rural roads, bank credit and economic growth

Dependent variable	$\Delta \log (\text{GDP} \dots)$		
	Overall (1)	Agriculture (2)	Services (3)
<i>A. Without controls</i>			
$\log(\text{Sum road}_{t-3,t-1}) \times \text{High rural credit}$	0.203** (0.086)	0.519** (0.248)	0.023 (0.104)
$\log(\text{Sum road}_{t-3,t-1}) \times \text{Low rural credit}$	-0.052 (0.122)	0.280 (0.327)	-0.072 (0.112)
Mean of dep. var.	7.32	6.25	6.54
Controls	No	No	No
District fixed effects	Yes	Yes	Yes
State $\times$ year fixed effects	Yes	Yes	Yes
Observations	2,766	2,766	2,766
<i>B. With controls</i>			
$\log(\text{Sum road}_{t-3,t-1}) \times \text{High rural credit}$	0.189** (0.087)	0.496* (0.253)	0.016 (0.106)
$\log(\text{Sum road}_{t-3,t-1}) \times \text{Low rural credit}$	-0.069 (0.121)	0.237 (0.332)	-0.078 (0.114)
Mean of dep. var.	7.32	6.25	6.54
Controls	Yes	Yes	Yes
District fixed effects	Yes	Yes	Yes
State $\times$ year fixed effects	Yes	Yes	Yes
Observations	2,766	2,766	2,766

(b) Roads, credit depth, and GDP growth

Figure 9: District-level credit and growth tests.

Read:

- Table 11 shows recent road construction predicts more rural private bank credit and deposits.
- A one-standard-deviation increase in roads is associated with roughly 10.5% more rural lending and 18% more rural deposits in the paper's interpretation.
- The same road variable does not generate comparable urban lending responses.
- Table 12 shows that roads are associated with higher GDP growth in high-rural-credit districts, especially agriculture.
- Low-rural-credit districts do not show the same response.

Economic message. Roads and credit are complements: infrastructure alone is more powerful when local finance is already able to respond.

6 Short summary

- The paper asks whether formal bank lending responds when infrastructure raises rural productive opportunities.
- It studies PMGSY, India's large rural road-building program.
- The empirical design uses population cutoffs in PMGSY road prioritization.
- Villages just above cutoffs are more likely to receive roads than nearby villages just below cutoffs.
- The core loan data come from a large private bank in Odisha and Uttarakhand.
- Density tests, maps, and balance plots support the validity of the cutoff design.
- Roads increase bank entry and the number of borrowers in a village.
- Roads increase loan amounts by about 30–35%.

- Loan maturity, default, and interest rates do not change meaningfully.
- The additional loans are concentrated in productive uses.
- Placebo tests using already connected villages find no effect.
- Heterogeneity tests show low-asset borrowers benefit disproportionately.
- District-level evidence shows more road construction is associated with more rural credit and deposits.
- Districts with deeper rural credit markets convert road construction into stronger output growth.
- The overall message is that infrastructure and finance work together in rural development.

7 Conclusion

7.1 Core conclusion

Agarwal, Mukherjee, and Naaraayanan show that private bank lending responds to improvements in rural productive opportunities. Villages just above PMGSY population thresholds receive more road access, and these villages see more borrowers and larger loans from the bank.

7.2 Economic intuition

Roads reduce transportation frictions and make market-oriented production more attractive. Villagers then have more profitable uses for borrowed funds, such as crop improvements, asset purchases, business expansion, and working capital. Formal banks respond by lending more, without charging higher rates or experiencing worse repayment.

7.3 Takeaway

The paper's main lesson is that rural finance does not operate independently of the real economy. Better roads create opportunities, and formal credit helps households exploit those opportunities. Infrastructure policy and financial development are therefore complements, not substitutes.